Unit 3 · Day 1

# Atom Discovery: Democritus to Dalton

Name: _____ Period: _____ Date: _____

## 1. Cold Open (before the sim)

Mr. D tears a piece of foil in half, again and again. Circle your prediction, then give a reason.

**I think a piece of matter can be cut in half forever / until I reach a piece that can't be cut**

Because _____

_____

Gold prediction (Part 1 of the sim): number of halvings from 1 cm to one atom = _____ Actual: _____

## 2. Sim Data (fill in as you go)

**Part 2, Make Water.** Record three different trials.

| Trial | H in (g) | O in (g) | Water (g) | Left over (what, how much) | Mass in = mass out? | O used ÷ H used |
|---|---|---|---|---|---|---|
| 1 | | | | | | |
| 2 | | | | | | |
| 3 | | | | | | |

**Part 3, Two Compounds.** Same hydrogen, different compounds.

| Compound | H used (g) | O used (g) | O per 1 g of H (g) |
|---|---|---|---|
| A (water) | | | |
| B (hydrogen peroxide) | | | |

O per 1 g H in B ÷ O per 1 g H in A = _____ Why a whole number? _____

## 3. Takeaways

**Two thinkers.** Fill the table.

| | Big idea | What backed it up? |
|---|---|---|
| Democritus (≈400 BCE)  | | |
| Dalton (1803)  | | |

**Dalton's five postulates.** Use the word bank. *Word bank: atoms · created · identical · rearranged · whole-number · subdivided*

1. All matter is composed of extremely small particles called _____.
2. Atoms cannot be _____, _____, or destroyed.
3. Atoms of a given element are _____ in size, mass and other properties. Atoms of different elements differ.
4. Atoms of different elements combine in simple _____ ratios to form compounds.
5. In chemical reactions, atoms are combined, separated, or _____.

**Three laws.** Name each one and give your example from the sim.

| Idea | Law (name it) | My evidence from the sim |
|---|---|---|
| Total mass before = total mass after | | |
| One compound always has the same mass recipe | | |
| Two compounds of the same elements: recipes differ by simple whole-number ratios | | |

**★ CHALLENGE TAKEAWAYS (if you got there)**

Dalton assumed water = HO. If that is true, one O atom is _____ times as heavy as one H atom. If water is $H_2O$, it is _____ times.

Nitrogen oxides, O per 14 g N: _____ g, _____ g, _____ g. Simple ratio: ___ : ___ : ___ Law: _____

Beyond Dalton. Something a solid ball can't explain: _____

_____

## 4. Exit Ticket

One thing Dalton's model explains, and one thing it can't. Use the sentence: "Dalton's model explains ____ because ____, but it can't explain ____."

_____

_____

| Rubric (10 points) | Points | What earns it |
|---|---|---|
| Cold open and gold prediction | 2 | Prediction circled with a reason; gold numbers filled in |
| Sim data tables | 2 | Trials and compound table filled with units |
| Two thinkers and five postulates | 3 | Table complete; postulates correct |
| Three laws and evidence | 2 | Each law named; each example comes from the sim |
| Exit ticket | 1 | Uses the sentence frame; one real explanation and one real limit |
| Stamp screen shown to Mr. D | — | Required to turn in the sheet. Challenge is extra, not required. |